

Streamlining Virtual KOL Generation Through Modular Generative AI Architecture

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Abstract

Key Opinion Leaders (KOLs) play a crucial role in modern marketing by shaping consumer perceptions and enhancing brand credibility. However, collaborating with human KOLs often involves high costs and logistical challenges. To address this, we present GenKOL, an interactive system that empowers marketing professionals to efficiently generate high-quality virtual KOL images using generative AI. GenKOL enables users to dynamically compose promotional visuals through an intuitive interface that integrates multiple AI capabilities, including garment generation, makeup transfer, background synthesis, and hair editing. These capabilities are implemented as modular, interchangeable services that can be deployed flexibly on local machines or in the cloud. This modular architecture ensures adaptability across diverse use cases and computational environments. Our system can significantly streamline the production of branded content, lowering costs and accelerating marketing workflows through scalable virtual KOL creation. Video demo is available at [here](#).

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Media arts**; • **Computing methodologies** → **Computer vision**; **Machine learning**.

Keywords

Modular Architecture, Generative AI, Image Generation

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1 Introduction

Key Opinion Leaders (KOL) are essential in marketing by improving brand credibility and shaping consumer perceptions. However, engaging top KOL can be costly and time consuming due to their high fees and content creation demands. To address this, businesses

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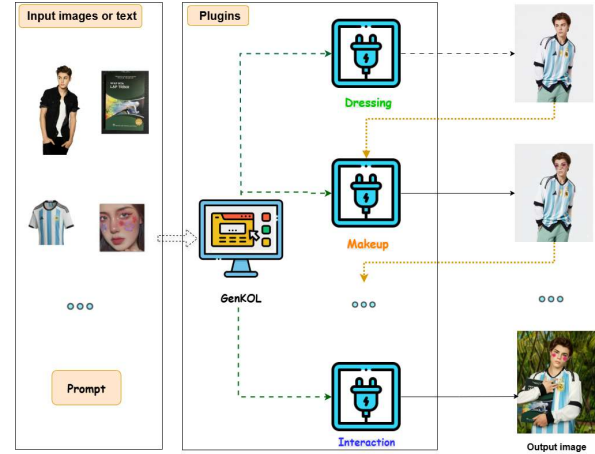


Figure 1: Workflow of Proposed GenKOL System.

can explore more cost-effective strategies. Advances in Generative AI [2] enable companies to automate promotional visuals, saving time and costs while broadening creative options.

This paper introduces a deep learning system for developing virtual KOL, called GenKOL, which offers an efficient and scalable way to promote products and services. Our system utilizes AI plugins to create customizable virtual KOL images for marketing content. It enables efficient resource management and scalable, modular workflows, allowing for easy integration of new AI capabilities without major changes. Our primary contribution is a flexible and modern architecture that facilitates the seamless integration of diverse Generative AI models [2] as independent, modular services. Each model, whether it be garment generation [4], makeup transfer [5], background synthesis [1, 3], or hair editing [6], is designed as a plugin, deployable in various runtime environments such as local machines, cloud servers, or edge devices. This plugin-based architecture enables the independent updating, replacement, or scaling of AI services based on user demands or available computational resources. By abstracting model functionalities behind standardized interfaces, the framework enhances reusability, accelerates development, and ensures cross-platform compatibility. This design reduces system coupling and accelerates the deployment and composition of AI pipelines, making it particularly well-suited for evolving most AI services. Video demo available at [here](#).

2 Proposed GenKOL System

2.1 Overview

Leveraging a flexible and modular AI plugin architecture (Figure 2), GenKOL enables users to generate context-rich virtual KOL visuals from input images and detailed text prompts while preserving

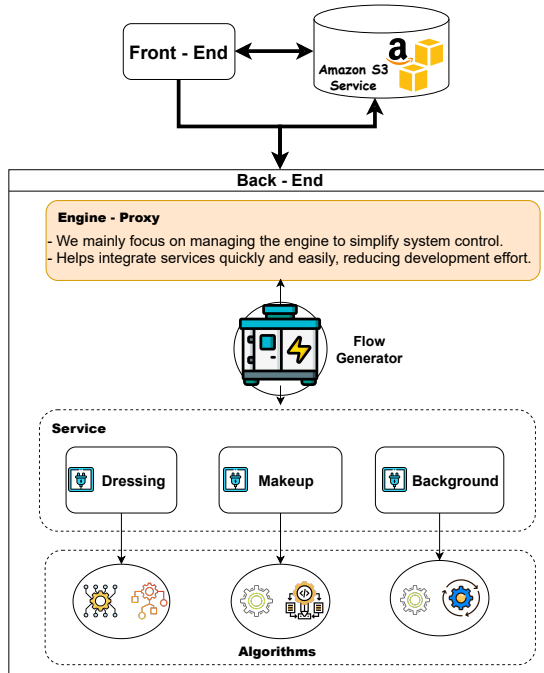


Figure 2: Proposed System Architecture

strong contextual coherence, outperforming traditional, loosely integrated pipelines. The system’s modularity supports scalable deployment, efficient resource use, and seamless integration of diverse generative models. By chaining specialized services into custom workflows, users can ensure smooth data flow, accelerate experimentation, and consistently produce high-quality, contextually aligned outputs.

2.2 System Architecture

We propose an architectural model designed to facilitate the integration of AI models as plugins. This architecture allows users to quickly and efficiently install, modify, and add services while conserving resources. Additionally, users can choose freely from the available algorithms, each with its own advantages and disadvantages. The architecture of our proposed system is illustrated in Figure 2.

2.2.1 Engine: The foundation of the architecture, this unified proxy layer standardizes communication among all AI services by defining a standard interface. Users can register AI services, deployed locally or in the cloud, by conforming to this interface, seamlessly integrating them as plugins. This modular design promotes interoperability, scalability, and ease of maintenance, allowing flexible composition of complex workflows and effortless updates without system disruption.

2.2.2 Flow Generator: The Flow Generator offers users a flexible way to dynamically create execution pipelines by combining various registered AI services. By utilizing the services integrated through the Engine component in section 2.2.1, users can easily define custom application workflows that meet their specific needs.



Figure 3: Our results with GenKOL.

This capability allows for the quick creation and adjustment of various sophisticated processing pipelines, fostering creative experimentation and enabling the efficient deployment of AI-driven applications.

2.2.3 Services - Algorithms: Registered services can host various algorithms, including AI models deployed locally or on cloud platforms like AWS or Google Cloud. This distributed approach enables users to flexibly utilize computational resources, optimizing resource use and reducing costs associated with computing power, memory, and GPU allocation while ensuring efficient and scalable execution of AI services.

2.3 Qualitative Results

To demonstrate the effectiveness of our proposed system, we provide visual examples in Figure 3 that illustrate the entire generation process. These examples show how specific inputs, such as a reference face image along with descriptive text prompts for clothing and makeup, are transformed into realistic virtual KOL images through a step-by-step execution of registered AI services. The resulting outputs exhibit high visual fidelity, with garments and facial features aligned with the given context. These examples highlight the system’s ability to maintain identity, incorporate various styling elements, and produce coherent, photorealistic results across different configurations.

3 Conclusion

In this work, we introduced a modular, plugin-based architecture for integrating multiple AI services, demonstrated through GenKOL, a system for generating realistic virtual KOL images. GenKOL offers flexibility, scalability, and ease of use, making it well-suited for AI-driven visual content creation in marketing. While effective, the system’s output quality depends on the performance and compatibility of individual plugins, and it currently lacks advanced scheduling and error-handling capabilities. Future work will focus on adaptive pipeline optimization, automatic plugin validation, and improved resource management to enhance reliability and expand applicability.

Acknowledgments

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